

Solution to the Penguin's Tour

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7×7 Grid

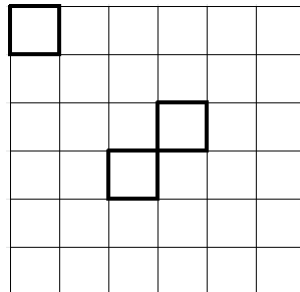
If we color the grid like a chessboard, such that dark squares are only adjacent to light squares and vice versa, we can see that every time a penguin moves, it goes from a square of one color to a square of the other color. This is because the penguin first moves 2 squares in one direction, which lands the penguin on a square of the same color, and then moves 3 squares in the other direction, which lands the penguin on a differently-colored square.

Since the tour alternates between light and dark squares, we can think of a penguin's tour as a polygon where each vertex is a square of the grid and each edge represents a penguin move. We can color each vertex of this polygon to match the color of its square. Since the penguin moves from a light square to a dark square and vice versa, each vertex must be the opposite color from its adjacent vertex. This means that there needs to be an equal number of light and dark squares.

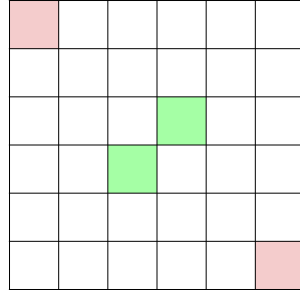
However, in a 7×7 grid, there are 49 squares. This means that it is mathematically impossible for there to be an equal number of light and dark squares, which means that a closed penguin's tour is impossible on a 7×7 grid.

6×6 Grid

On a 6×6 grid, the penguin runs into a problem along its tour: it has no way to access all the corners of the grid! We can see that each corner corresponds to two squares from which the penguin can access that corner:



However, if we look at a pair of diagonally opposing corners, we notice that the access squares for both corners are the same:



Since this coloring of the grid is symmetric, we can say that the other two corners in the diagram are red, and the other two central squares are therefore also green. Using this method of sectioning off the board, we can create three separate cases for the penguin's initial square:

Case 1: Initial square is a corner (red square)

If the penguin starts on a red square, it has to exit onto one of two available green squares. Since the penguin is now on top of a green square, it must go to the only available red square, or else it won't be able to exit that corner later without landing on the same green square twice.

However, once the penguin enters the corner, it must exit onto the only available green square. Since the penguin then has to go to a white square in order to land on all the other squares, it has to leave the access square, which means that it has no avenue back to its initial square, which violates the conditions of a closed penguin's tour and therefore makes the tour impossible.

Case 2: Initial square is an access square (green square)

If the penguin instead starts on a green square, it immediately needs to go to a corner square, since going to a non-corner square will prevent the penguin from reaching both corners because the penguin is unable to re-enter its initial square except to close the tour, therefore preventing it from reaching both corners.

Since the penguin needs to go to a corner, it then has to exit through the other access square. In order to reach the diagonally-opposite corner, it has to go to the corner immediately after entering that access square, since not doing so would lock off that corner. However, upon entering the corner, both access squares have already been used, resulting in the penguin being unable to exit, which means that regardless of what happens, the penguin is unable to reach every square on the board if it starts on an access square.

Case 3: Initial square is neither an access square nor a corner (white square)

If the penguin starts on a white square, then the only way for it to get to a corner square is to get to an access square first. Going to an access square without reaching any corners first has already been proven in Case 2 to not work, which means that starting on a white square is futile.